

Hang Yin

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Education

09/2023-08/2024	MSc. in Statistics, University College Dublin, GPA: 3.73/4.0 (First Class Honours) Dissertation: <i>Monte Carlo Simulation of RAFT Polymerization</i> Core Modules: Monte Carlo Inference, Stat Network Analysis, Mathematical Statistics, Time Series Analysis, Advanced Predictive Analytics, etc.	Ireland
09/2019-06/2023	BSc. in Statistics, Beijing Jiaotong University, GPA: 3.16/4.0 Dissertation: <i>Research on Spin Glass System based on Ising Model</i> Core Modules: Statistical Analysis of MATLAB, Data Analysis(R), Multivariate Statistical Analysis, Statistical Computation, Physics, etc.	China

Professional Experience

10/2023-01/2025	Visiting Student/Researcher, Charles Institute of Dermatology, • Developed Monte Carlo simulation models to study polymerization mechanisms and kinetics. • Enhanced simulation efficiency using algorithm optimization techniques. Group Website: https://www.wenxinwang.group/staff.html	Ireland
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Publications

[1] 2025 (Accepted)	Yin H. , Li Y., and Lyu J., Hybrid Stochastic Simulation for Accelerated Modeling of Free Radical and Degenerative Transfer Polymerization. <i>Macromolecular Theory and Simulations</i> . Early version preprint: https://doi.org/10.26434/chemrxiv-2025-gxj4r
[2] 2025 May	Li Y., Yin H. , Yi S., Lyu J., and Wang W., Beyond Flory's Principle: Cyclization and Unequal Reactivity in Step-Growth Polymerization. <i>Science Advances</i> . https://doi.org/10.1126/sciadv.adu8884

Research Experience

05/2024-12/2024	Monte Carlo Simulation of RAFT Polymerization (MSc Dissertation) This project aimed to investigate the necessity of intermediate states in RAFT polymerization: • Simulated FRP, DT, and RAFT polymerization systems using standard SSA; • Developed a hybrid algorithm that significantly enhanced simulation efficiency; • Verified the performance and scalability of the new algorithm across different system sizes; • Programming languages: R, Julia; Outcome: [1].
10/2023-05/2024	Cyclization and Unequal Reactivity in Step-Growth Linear Polymerization Refined Flory's principle and introduced new mathematical formulations for Step-growth polymerization with cyclization and unequal reactivity: • Applied Monte Carlo simulations to validate and improve theoretical models; • Proposed a simulated annealing approach to estimate cyclization deviation; • Programming languages: MATLAB, R; Outcome: [2].
12/2022-06/2023	Research on Spin Glass System based on Ising Model (BSc Dissertation)

Explored spin glass behavior using the Ising model under both classical and quantum annealing frameworks:

- Simulated annealing dynamics of the Ising model and investigated hysteresis under magnetic fields;
- Applied both quantum and classical annealing to the Edwards-Anderson model and compared performance;
- Programming language: MATLAB.

Skills & Interests

Programming Languages:	R, Python, MATLAB, Julia, C++, SQL.
Statistical Methods:	Machine Learning, Predictive Modeling, Time Series Analysis, Monte Carlo Simulation, Statistical Inference.
Software & Tools:	Excel, Power BI, SPSS, PostgreSQL, Git.
Specialized:	Statistical Modelling, Monte Carlo Simulation, Algorithm Development.
Interests	Fitness training, Open water diving, Amateur astronomy.

Award

2021	Learning progress scholarship of Beijing Jiaotong University
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Extracurricular Roles

Member of the UCD Math Society;
Member of the UCD Physics Society;
Vice Director of BJTU Science and Technology Association;
Member of the Study Section of the BJTU Students Union;
Member of the Astronomy Club of BJTU;
Beijing Metro Volunteers.
